

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 2

Error Analysis Task

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

FACULTY: Dr. D.SUDHAGAR

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 30%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I NALIPI REDDY ROHITH/192425115 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:N.Rohith

QUESTIONS: 05

Total Marks:25

1. Error Analysis in Maximum Likelihood Estimation (MLE)

a) Conceptual Error

The likelihood was written as the **sum of probabilities** instead of the **product of probabilities** for independent observations.

b) Effect on Parameter Estimation

- Produces an incorrect objective function.
- Parameters no longer maximize the true likelihood.
- Results in biased and inaccurate estimates, even with large datasets.

c) Correct Formulation

For independent samples:

$$L(\theta) = \prod_{i=1}^n P(x_i | \theta)$$

The parameter estimate is:

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

This correctly represents the joint probability of observing all samples.

d) Role of Log-Likelihood

Log-likelihood is:

$$\log L(\theta) = \sum_{i=1}^n \log P(x_i | \theta)$$

Advantages:

- Converts products into sums.
- Prevents numerical underflow.
- Simplifies differentiation and optimization.

2. Error Diagnosis in Perceptron Learning

a) Possible Implementation Errors

- Incorrect weight update formula.
- Missing bias update.
- Wrong prediction rule.
- Learning rate set improperly.
- Data labels not encoded as ± 1 .

b) Effect of Learning Rate or Weight Updates

- Very large learning rate causes oscillation.
- Very small learning rate slows convergence.
- Incorrect updates prevent the weights from reaching the correct boundary.

c) Dataset-Related Issues

- Incorrect labels.
- Features not normalized.
- Dataset is actually not linearly separable.
- Presence of noisy or duplicate samples.

d) Modifications

- Normalize features.
- Use correct Perceptron update rule.
- Include bias term.
- Choose an appropriate learning rate.
- Verify that the dataset is linearly separable.

3. Backpropagation Gradient Error Analysis

a) Possible Errors

- Incorrect derivative of activation function.
- Wrong chain rule implementation.
- Incorrect sign in gradient update.
- Learning rate too large.
- Programming mistakes in matrix operations.

b) Role of Activation Functions

Activation functions determine gradient flow.

- **Sigmoid/Tanh:** Can produce very small gradients.
- **ReLU:** Reduces vanishing gradients for positive inputs.

c) Vanishing and Exploding Gradients

- **Vanishing Gradient:** Gradients become near zero, slowing learning.
- **Exploding Gradient:** Gradients become extremely large, causing unstable updates and increasing loss.

d) Techniques to Fix

- Xavier or He weight initialization.
- Batch Normalization.
- Gradient Clipping.
- Use ReLU activation.
- Reduce learning rate.

4. Gradient Descent vs SGD Error Investigation

a) Why Batch Gradient Descent is Slow

- Uses the entire dataset for each update.
- High computational cost for large datasets.
- Longer time per iteration.

b) Cause of SGD Fluctuations

- Updates parameters using one sample at a time.
- Gradient estimates are noisy.
- Loss fluctuates around the optimum.

c) Mini-Batch Gradient Descent

- Uses small batches (e.g., 32–256 samples).
- Faster than Batch Gradient Descent.
- More stable than SGD.
- Provides smoother convergence.

d) Best Approach

Mini-Batch Gradient Descent is preferred because it:

- Balances speed and stability.
- Reduces noise in updates.
- Converges efficiently on large datasets.

5. Curse of Dimensionality Error Analysis

a) Effect of High Dimensionality

- Feature space becomes sparse.
- More training data is required.
- Model memorizes training data instead of learning patterns.

- Generalization error increases.

b) Signs of Overfitting

- High training accuracy.
- Low testing accuracy.
- Large gap between training and testing errors.

c) Why Distance-Based Learning Fails

- Distances between data points become nearly equal.
- Nearest neighbors become less meaningful.
- Similarity measures lose effectiveness.

d) Techniques to Improve Performance

- **Dimensionality Reduction:** PCA, LDA, Autoencoders.
- **Feature Selection:** Remove irrelevant features.
- **Regularization:** L1 (Lasso), L2 (Ridge), Dropout.
- **Cross-Validation:** Improve model generalization.

Conclusion: Reducing dimensionality and applying regularization decreases overfitting and improves test performance.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand